SYMPHONY
NEWSLETTER OF THE THIRUVANANTHAPURAM CHAPTER
OF THE COMPUTER SOCIETY OF INDIA
June 2023 – May 2024
Vol.45

CSI-InApp International Student Project Awards 2023

The 12th edition of the CSI-InApp International Student Project Awards was launched on 24th January 2023. InApp has been organizing these awards in association with CSI (Computer Society of India), Trivandrum Chapter for Bachelor's Degree students of Computer Engineering starting from 2012. The award is initiated to recognize and promote students who demonstrate exceptional talent in developing innovative computer applications as part of their final year project.

The best project was awarded a cash prizes of Rs. 50,000 followed by two consolation prizes of Rs 25,000 each and an award plaque. This year, more than 2300 students participated from 10 states of India. It was an arduous and challenging task for the jurists to evaluate 600 projects. Kerala Technological University joined as the Academic Partner. After two rounds of screening and shortlisting, and a final round of presentations, the best project award was bagged by Kolhapur Institute of Technology, College of Engineering, Maharashtra for their work on Intelligent Sugarcane Eye Bud Cutter for nurseries. The consolation prizes went to K.J. Somaiya Institute of Technology, Maharashtra for their project KalMiFyn , a self-help app facilitating the mentally depressed and P. D. A. College of Engineering, Karnataka for their cyber-security software titled Secure Digital identity using NFTs.

The awards ceremony was held on 12th October, 2023 at Hotel Gokulam Park, Aakulam, Trivandrum. Dr. Saji Gopinath, Vice-Chancellor of Digital University, Kerala was the Chief Guest. Members present from the CSI-InApp team were Mr. Vijaykumar, CEO InApp, Dr. Sasikumaran Sreedharan, Chairman CSI Trivandrum Chapter and Ms. Soja Anil, Coordinator InApp team. Sree Kumar V., Centre Head, Tata Elxsi was a special invitee. The occasion was graced by dignitaries from industry and academia. Congratulations to the winners who came all the way from different states to make the programme a grand success.





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Editorial Advisory Board

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Dr. Tessy Mathew
MC Member & SAC

Rajesh P
Vice-Chair

Editor.Write( )

I am profoundly honored to serve as the Editor-in-Chief of Symphony, Newsletter of CSI Trivandrum Chapter during my tenure as Chairman for the period 2023-2024.I take this opportunity to express my thanks to the Editorial team under the leadership of Dr. Elizabeth Jacob for their active role in compiling the contents of this prestigious Newsletter. The programmes organized by SAC are also well planned and arranged by Prof Tessy Mathew. I also place on record the extensive support given by the Management Committee and the Election Committee for discharging various responsibilities as per the objectives of CSI.

As in the previous years, the chapter had to suffer due to the lack of chapter share of funds from the CSI headquarters. Though we have initiated the conduct of two training programmes on new technology areas, namely, cyber security and AI & Machine Learning tools & applications, there were only few enrolments. We have also initiated internship training programs for the students of colleges having CSI student chapters in association with a popular Chennai based PAN India company uLektz Learning Solutions Private Limited, however, due to shortage of time, it could not be implemented. The new committee may continue this initiative so that CSI will be able to add more student chapters and institutional memberships.

We have taken steps to set off long pending issues such as revising the MOU with InApp in connection with CSI-InApp International Student Project Awards, completing resurvey procedures for land and building tax payments, discontinuing services of supporting staff and completing electricity phase conversion process. The efforts of Rajesh, Vice Chairman in settling these issues are well appreciated.

The role of professional bodies is becoming more significant than ever before in the changing scenario of technology transformations resulting in a large contemporary technology gap between the professional graduates passing out from universities in India and requirements of industry. The growing unemployment among professionals and other graduates are the pointers to show the gravity of the prevailing situation and the urgency to take up measures to alleviate this problem by skilling the huge Indian manpower. The new education policy released in 2020 suggests many alternatives to produce employable professionals to address these issues by giving academic autonomous powers to educational institutions and allowing them to collaborate with industry and also with top 500 ranking international universities. The demand for new technology enriched solutions are also expected from various sectors of economy. The role of CSI, having members with variety of expertise, can do a lot in bridging the technology gaps.

There are many technologies emerging in the recent years that are creating large-scale job opportunities. Latest additions to these technologies are Virtual production,3-D modeling, 3-D Technologies and Art, Metaverse related technologies such as AVGC – XR (Animation,Visual effects,digital design,Gaming

An appeal to all members

The past MCs have made concerted efforts to curate the member database. In continuation, all new members and members who are yet to update their contact information are requested to mail their present position (or affiliation before retirement), email, mobile number to csitvm@gmail.com. This would help in updating the electoral rolls and also to network within our community.

Editor.Write( ) *Contd. From Page 2*

and Virtual Reality) technologies powered with Gen AI Tools. AI can provide an INR 450 billion boost to the Indian Media & Entertainment(M&E) sector by 2027.

AVGC Technologies are identified by both central and state governments as great opportunity technologies for generating large-scale employment/self-employment irrespective of one's qualifications & basic background. Indian M&E industries are expected to grow and reach INR 2.5 Trillion by 2024 and grow at a CAGR of 10% to reach INR 3.08 Trillion by 2026 based on FICCI & EY 2024 reports. To augment the activities related to these technologies, Media & Entertainment policies have also been released by government. It is also considered as a futuristic technology for harnessing start-up opportunities.

I am sure that CSI shall play a vital role in organizing programmes, particularly to impart advanced technology skills and other supporting skills on Adaptability, Emotional Intelligence, Resilience power, leadership and interview skills to equip the unemployed youth. CSI shall also function as an interface between educational institutions, industry and youth. There is a great opportunity to fulfil the mandate of CSI. Hope the new committee will take appropriate steps to continue the activities initiated by the outgoing committee and also conduct new programmes to serve the community. I take this opportunity to convey my regards to each and every member of the Management Committee for their excellent support and cooperation to perform various activities of the chapter.

Thanking You
Prof. Dr. Sasikumaran Sreedharan
Chairman, CSI Trivandrum Chapter

Technical Review of CSI-InApp Award Winning Projects

Best Project - Intelligent Sugarcane Eye Bud Cutter for Nurseries

Kolhapur Institute of Technology (KIT) College of Engineering, Maharashtra

Team Members - Yash Bhalchandra Petkar, Prerna Vijay Rawale, Ravi Rajesh Shah and Adesh Ashok Ingale

Sugarcane stalks are cut manually using machetes and tools resulting in uneven cutting, wastage and injuries to labourers. An eye bud cutter can, not only improve the working conditions of the labourers but also address the shortage of skilled labour in the sugarcane industry by automating a repetitive and physically demanding task.

The intelligent cutter can precisely locate and cut the eye buds on the stalks using cameras, advanced sensors and software. Machine learning algorithms (yolov5) detect the location and size of the eye buds on the sugarcane stalks. The blade of the cutter is powered by a motor and controlled by a YOLO that will determine the optimal cutting path for each stalk. The technology is built on Raspberry PI 4B using Python language.

The cutter would enhance productivity by improving the quality and yield of the crop and making the harvesting safe and efficient. A functional prototype has been demonstrated.



Technical Review of CSI-InApp Award Winning Projects

Consolation Prize : KalMiFyn (Read: CallMeFine)

K.J. Somaiya Institute of Technology, Maharashtra

Team Members - Abhishek Khandge, Tanishk Jain, Rhutuja Vispute and Richa Bhandari

KalMiFyn is a mobile app that leverages dependable data to provide companionship and effective individualized mental health care while keeping anonymity and data security. The project aims to address data breaches, data security, individual users› mental health needs, and the stigma associated with seeking professional help by developing emotionally intelligent and socially compatible AI. In the Voice Authentication stage users are authenticated on the basis of their voice features. In the next Avatar stage, KalMiFyn takes voice queries from the user, based on the input data, the AI model generates possible solutions which are converted from text to audio, for a contactless experience. The Avatar ecosystem is developed under the supervision of clinical Psychiatrists and Psychologists in collaboration with K.J. Somaiya Medical College and Research Centre.

Consolation Prize : Secure Digital identity using NFTs

P. D. A. College of Engineering, Kalaburagi, Karnataka

Team Members - Venkatesh Patil, Spoorti S Patil and Syed Yaseen Raza

In today's digital era, personal identity information is vulnerable to breaches and cyber attacks due to centralized storage on servers. Users lack control over their identity information, leading to the potential misuse of their personal information. To address this issue, a digital identity solution has been developed using Non-Fungible Tokens (NFTs), which gives users full control over their personal information.

This could involve providing personal information, such as name, date of birth, address and Aadhaar number to verify the accuracy of the information. Once the individual's identity has been verified, they would be issued a unique NFT (representing their identity) which is stored in their Metamask wallet (or any compatible wallet). This NFT can be used to verify user's identity on various web3 platforms. Blockchain technology is used for creating secure digital identities with NFTs and the system is built on the Truffle platform.

Office bearers for 2023-24



Sasikumaran Sreedharan
Chairperson



Rajesh P.
Vice-Chair



Lekshmy Money
Secretary



Justin G. Russel
Treasurer



Sajeendran S
Secretary, Adhoc



Elizabeth Jacob
Member



Tessy Mathew
Member



Aju Thomas
Member

The office bearers took charge in the 44th AGM held on 10th June, 2023 at the Chapter Office. The Nomination Committee members for 2023-24 are :

1. **Achuthsankar S Nair** (Chairman)
2. **R P Nair**
3. **Kesavan Nair**

FC Kohli National Tech Talk Series

In homage to the former CSI president and IT visionary, CSI Trivandrum Chapter conducted the FC Kohli Commemorative National Tech Talk Series as a webinar series where students could talk on emerging technologies of their interest. It also provides a platform for young, enthusiastic minds to present their project and research activities. The Colleges which participated are Mar Baselios College of Engineering and Technology, LBS Institute of Technology, Trivandrum, College of Engineering Perumon, SCT College of Engineering Trivandrum, GEC Barton Hill, TKM College of Engineering, Quilon, Christ Nagar College, Mohandas College of Engineering and Technology and Sree Buddha College of Engineering.

The first of the series held on January 31, 2024, was hosted by the College of Engineering, Perumon and saw active participation from 67 students. Dr. Biji, Associate Professor at VIT Vellore, chaired the session. Dr. Biji's expertise and insights provided valuable context and depth to the discussions, enhancing the overall quality of the webinar.



Topics presented

WESAT : An informative talk on the country's first Women Engineered SATellite, developed by an all-women's team at LBS Institute of Technology for Women, Trivandrum. The small satellite weighing 1 kg was launched aboard the PSLV-C58 mission as a co-passenger satellite. It measures UV rays on the earth's surface to study their influence on climate change in Kerala. WESAT live data is available from the ground monitoring station in Trivandrum.

GENAI in Big Data: Discussion on the integration of Generative AI in Big Data analytics, highlighting how AI is revolutionizing data processing and decision-making.

ChatGPT and Generative AI: An in-depth look at ChatGPT and other generative AI models, exploring their applications, potential, and implications in various industries.



The webinar held on February 16, 2024, was hosted by the Government Engineering College, Barton Hill and witnessed active participation from 30 students. The session was chaired by Dr. Elizabeth Sherly, Professor Digital University.

Topics presented:

Spiking Neural Networks: advancements and applications of spiking neural networks, an ANN that mimics the operation of biological neurons.

Metaverse: Discussion on the development and implications of the metaverse, exploring its potential to revolutionize social interaction, commerce, and entertainment.

How Far Can Large Language Models Be Optimized: An in-depth look at the optimization of large language models, examining the current state of the art and potential future advancements in this rapidly evolving field.

The webinar held on April 18, 2024 was hosted by the Mar Baselios College of Engineering and Technology. 35 students participated. The session was chaired by Dr. Achuthsankar S Nair, Professor Digital University.

Topics presented:

Quantum Computing: A presentation on the advancements and applications of quantum computing, exploring its potential to revolutionize computing power and problem-solving capabilities.

Object Detection and Tracking System for the Visually Impaired: Discussion on a system designed to aid visually impaired individuals by detecting and tracking objects in their environment, enhancing their ability to navigate and interact with the world.

Virtual Assistant App for Visually Impaired: An in-depth look at a virtual assistant application tailored for visually impaired users, providing functionalities that support daily activities and improve accessibility.

Dr. Tessy Mathew, SAC (Student Activities Committee) Coordinator worked with different student branches to conduct the webinar series.

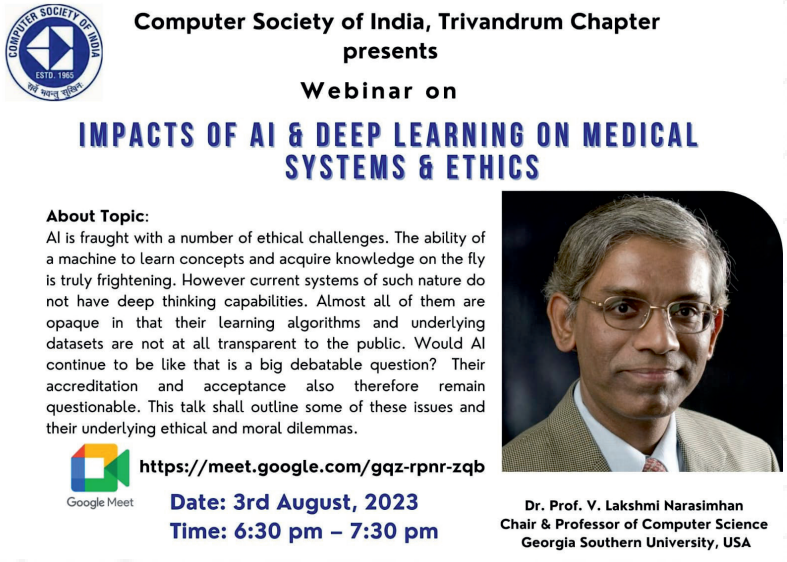
Institution membership was renewed by Christ Nagar College and Mohandas College of Engineering and Technology



Summary of Activities in 2023-24

Webinar by Prof. V. Lakshmi Narasimhan, Georgia Southern University, USA

The webinar hosted by CSI, Trivandrum Chapter on 3rd August 2023 about "Impacts of AI and deep learning on medical systems and ethics" was thought provoking and laid bare the ethical and moral challenges of this technology as AI races against its own makers.



New MoU with InApp inked for CSI-InApp Student Project Awards

After many discussions, a formal MoU was signed between CSI Trivandrum Chapter and InApp Technologies Pvt Ltd in January 2024. All award related matters like prize money announcements, project evaluation, IPR issues will be decided by a coordination committee consisting of members from each side. The MoU takes effect from 2024 onwards.

National Student Tech Talk Series conducted in January, February and April 2024.

Course offered on AI/ML Tools & Applications

The 100 hours online certificate course covers fundamentals of AI and Machine Learning, programming with Python, data analysis and visualization, Deep learning, Generative AI & Conversational Agents, practical application development with extended follow up support. Successful completion of the course would make participants employable in job roles like Data Analyst, Data Scientist, Machine Learning Engineer, Data Engineer and Business Intelligence Analyst. In addition, the course can upskill professionals who need to adopt AI and integrate these technologies into their domain areas. The Instructor is Er. P.G. Muralidharan. The course is nearing completion.

Computer Society of India

CERTIFICATE IN AI & MACHINE LEARNING TOOLS & APPLICATIONS (CAIML)

ONLINE INSTRUCTOR LED TRAINING

Introduction:
In an era dominated by data, AI & ML have emerged as game-changers, revolutionizing the way we live, work, and interact. As industries increasingly integrate these technologies, individuals armed with AI & ML expertise become indispensable. The significance of the Certificate in AI & Machine Learning Tools & Applications (CAIML) lies in its ability to equip participants with the skills needed to navigate this data-driven landscape confidently. The program not only ensures you stay ahead of the curve but positions you as a catalyst for innovation, ready to tackle the challenges of the future.

Highlights & Duration:
- Immersive Learning Experience
- Interactive workshops and labs
- Industry Expert-led sessions
- 100 percent tool-oriented training with concepts to make participants a job ready AI & ML professional.
- Case studies and industry insights
- Participants are required to use their own Laptops.

Duration: Total 100 hours @ 2 hours per day (Tuesday & Thursday 7 to 9 PM) and 3 hours per day (Saturday & Sunday 2 to 5 PM)

Course Contents:
Fundamentals of AI and Machine Learning, AI Programming with Python, Data Analysis and Visualization, Deep learning, Generative AI & Conversational Agents, Practical application development, Extended follow up support. Certificates will be provided to participants.

Pedagogy: Theoretical sessions and hands on exercises.

Date & Time:
The training program is tentatively scheduled during second week of February 2024.

Who can Attend:
- Degree/Diploma in Engineering (completed/doing) - Fresh/Experienced.
- BCA, MCA, BSc CS, MSc CS, PGDCA (completed/doing) - Fresh/Experienced.
- Other Graduates - Fresh/Experienced with programming skills desirable

Job roles, Employable organizations & Standard salary packages
Job roles: Data Analyst, Data Scientist, Machine Learning Engineer, Data Engineer, Business Intelligence Analyst.
Employable Organizations: All Types of Industries, Banking, IT Companies, Health departments &Systems, Defense, Police, Govt Departments, E Governance, Insurance, Etc.
Standard salary range: Average starting salary for AI engineer with no prior experience in India is around ₹3.0 to ₹5.0lakh per year and experienced professionals can get a job more than₹6.0 lakh per annum.

Course Fees & Registration:
- CSI/IEEE Student Members - **Rs. 25,500/-** + GST @18% per participant
- CSI/IEEE Members - **Rs. 26,500/-** + GST @18% per participant
- General Public- **Rs. 27,500/-** + GST @18% per participant

All payments should be made by DD/ Cheque in favor of **"CSI Trivandrum Chapter"** Payable at Thiruvananthapuram or through Net Banking in two equal installments (50% at the time of registration and 50% in two weeks after commencement of the training). Subject to Trivandrum Jurisdiction.

CSI Account details. Account Name: CSI Trivandrum Chapter; Account Number: **33853741841;** Bank: State Bank of India; Branch: MIDC Andheri (East), Mumbai; Branch Code: 07074; Type of Account: Savings Bank; IFSC Code: SBIN0007074

For registration, please visit **register.csitvm.org**

Contact Address:
Computer Society of India Trivandrum Chapter, Reg. No. 41/1965, T.C. 82/5426, Symphony, Cherian's Square, Ambujavilasam Road, Thiruvananthapuram, Kerala, India, PIN: 695001
Email: csitvm@gmail.com, www.csitvm.org, fb.com/csitvm
Mobile / WhatsApp: 9400796741 GSTIN: 27AAATC1710F1ZC PAN: AAATC1710F
(Batch size: Maximum 25-30 participants only. Please confirm your Registration by Phone or Email.)

Student internships offered by ULektz Learning Solutions Private Limited

uLektz Learning Solutions Private Limited (uLektz) is an ISO 9001 and ISO 27001 certified edtech company headquartered in Chennai. Positioned as #BharatKaLinkedIn, they aim to bridge the skill gap between academia and industry. The Company had submitted a proposal for collaboration with CSI, Trivandrum Chapter to support students and institutions of higher education across the state of Kerala with industry connect, skill development, internships and job placements. The implementation plan worked out was to provide CSI Trivandrum Chapter with the uLektz branded mobile app. CSI can add/invite student chapters of institutional members to download and use the mobile App for networking within the college, enrol for short-term internships (as per UGC directives for personal development and enhancing employability) and also get access to vast resources for learning. It will bring more institutions of higher education under the purview of CSI. The student community stands to benefit immensely from this state-wide outreach programme. Discussions are ongoing with institutions and it is hoped that the next committee will take it forward to fruition.



Administrative Matters

Austerity measures were adopted to curtail the recurring expenditure of the Chapter. It was decided by the MC to meet online, unless necessary. The service of the office assistant would be sought on a need basis. In order to save on energy bills, the process of converting to single phase connection is almost completed. The BSNL landline was surrendered, a prepaid Jio connection taken and UPI payment option has been enabled for ease of carrying out transactions.

The Chapter office is housed in Cherian Square Building, located in the heart of the city. Mr. Rajesh P, Vice-Chair took the initiative to address property related issues that needed liaisoning with government departments and service providers. Ownership certificate was obtained from the Corporation after paying the property tax dues. The re-survey process has been completed and land tax arrears cleared from the village office.

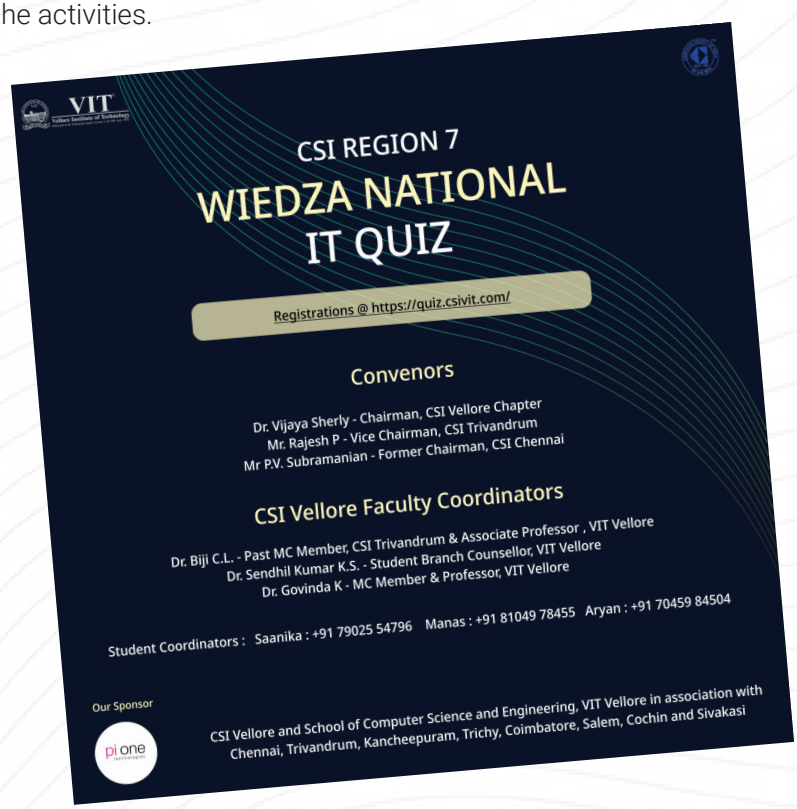
Ongoing Acitivities



CSI-InApp International Student Project Awards 2024 edition has been launched with cash prizes and placement opportunity for all finalists. The first round review of projects by faculty members, is in progress. From CSI, Dr. Tessy Mathew and Rajesh P. are coordinating the activities.

Wiedza National IT Quiz 2024

CSI Vellore and School of Computer Science and Engineering, VIT Vellore, in association with the other Chapters of Region 7, is conducting this year's Wiedza National IT quiz. The event encourages students to pep up their knowledge in theory and applications of computer science and technology. The competition is open to all graduate/postgraduate students studying in colleges across the country. Mr. Rajesh P, Vice Chairman, CSI Trivandrum is one of the Convenors.



Remembering J.T. George, a Founding Member of our Chapter



John T. George, Scientist VSSC, former Chairman (1980-81), and a Chapter Patron of CSI, Trivandrum, passed away in May 2024 at the age of 88. He began his career in 1957 in the Nuclear Instrumentation Group at the TIFR, Mumbai. In the early 60s, as MINSK, the first computer of Trivandrum made its way from the Soviet Union to VSSC, he was assigned the task of establishing and managing the mainframe computing facility at the Thumba Equatorial Rocket Launching Station (TERLS). Along with his colleagues, he organized the Annual Convention of CSI which was held in Trivandrum for the first time in 1969.

In more than three decades of service, he headed the Rocket Systems Testing Division and later was Head, Quality Systems Reliability at VSSC. He was an active member of our Chapter till recently. In 2021, at the initiative of our Chapter, he handed over the relics of the MINSK machine to the Computer Gallery of Kerala State Science and Technology Museum. The Chapter pays rich tributes, recognizing his life-long contributions. Our condolences to the bereaved family.

AGM 2023







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